Ghee Boiler



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Ghee Boiler

What it is used for

To heat cream or butter in a ghee boiler to remove moisture and develop the desired ghee characteristics. To control heating, monitor the product, and maintain hygienic operation.

Main parts

- Heating vessel or boiler
- Heating element or burner
- Temperature controller and sensor
- Agitator, if fitted
- Lid and outlet valve
- Insulation and control panel

Before you start

1. Ensure the vessel is clean, dry, and suitable for food processing.
2. Inspect the heating source, wiring or fuel connections, temperature sensor, and controls.
3. Check that the outlet valve is closed and the lid and seals are in good condition.
4. Confirm that the boiler is installed on a stable surface with suitable ventilation.
5. Prepare the required quantity of cream or butter and avoid exceeding the marked capacity.
6. Wear suitable heat-resistant gloves, protective clothing, and closed footwear.

How to use

1. Switch on the heating system according to the manufacturer's instructions.
2. Load the prepared cream or butter without exceeding the vessel's capacity.
3. Set the temperature and heating mode specified by the approved process.
4. Start agitation if fitted and required by the process.
5. Heat gradually while monitoring temperature, product appearance, and aroma.
6. Adjust the heat as required to prevent scorching, foaming, or overflow.
7. Continue processing until the product reaches the approved moisture-removal and ghee-finish criteria.
8. Switch off the heat before unloading or opening outlets, following the equipment procedure.
9. Allow the product to settle as required and drain through the designated outlet into a clean, dry container.
10. Filter the ghee using the approved process if filtration is required.
11. Switch off and isolate the power or fuel supply after the operation.
12. Clean the vessel and product-contact parts using the approved food-safe cleaning procedure.

⚠ Safety

- Only trained personnel should operate the boiler.
- Never exceed the manufacturer's fill limit or operating temperature and pressure limits.
- Avoid contact with hot vessel surfaces, hot ghee, steam, and splashes.
- Do not open a pressurized vessel; confirm that pressure is safely released if pressure operation applies.
- Keep water away from hot fat to prevent dangerous splashing.
- For gas-fired units, check for fuel leaks and maintain adequate ventilation.

Care & storage

- Clean and dry all product-contact surfaces after use.
- Inspect seals, valves, agitator, and temperature sensor regularly.
- Remove product residue before it hardens using an approved cleaning method.
- Check electrical connections or burner components as applicable.
- Arrange periodic maintenance and temperature-control checks.

Common mistakes to avoid

- Overfilling the vessel and allowing the product to overflow.
- Heating too rapidly and scorching the product.
- Ignoring temperature readings or unusual smells.
- Allowing water or cleaning chemicals to contaminate finished ghee.
- Cleaning or servicing the unit before it has cooled and been safely isolated.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar